

# Improving Explainability in BERT-Based NLP Models Using Faithfulness Regularization

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April 4, 2026

## Abstract

Transformer-based language models have shown impressive performances on a variety of natural language processing problems; yet, they exhibit opaque decision-making processes. Current approaches to explaining black-box models focus on attribution techniques, which emphasize critical tokens without providing evidence of their influence on the model’s output. While explanations produced by such techniques may be convincing, they usually fail to meet faithfulness standards and cannot be trusted in real-world applications. Therefore, we propose a faithfulness-aware approach to generating explanations, which includes perturbation-based regularization in training language models. In particular, our method introduces faithfulness-oriented penalties that perturb highly influential tokens and motivate the model to make significant changes in its predictions, thus increasing consistency between selected tokens and actual model behavior. Our framework estimates token importance using gradient-based attribution and integrates it into a rationale-dependent training procedure. We demonstrate the effectiveness of our approach on various NLP problems such as sentiment classification, natural language inference, and hate speech identification. Experiments involve faithfulness-focused evaluation techniques, e.g., token deletion tests. We show that incorporating faithfulness-related regularization objectives during training improves the explainability of BERT-based NLP architectures without sacrificing their performance on target tasks.

## 1 Introduction

There has been tremendous success achieved in the field of natural language processing (NLP) through the development of language models such as BERT based on transformers which have demonstrated state-of-the-art performance across multiple tasks like sentiment analysis, question answering, text classification, and natural language inference. This success can be attributed to the ability of such language models to make use of deep learning methods, specifically self-attention neural networks, which allow the model to learn contextual dependencies spanning long ranges of textual input. This is what makes transformer-based language models the prevailing architecture for contemporary NLP models both in industry and academia. On the downside, such language models cannot provide any meaningful insight into how predictions are made since their inner workings are quite opaque due to complex layers of attention networks and large parameter spaces.

In applications where interpretability is crucial for accountability and transparency, the uninterpretability of machine learning algorithms becomes a key drawback. In fields including healthcare decision-making assistance, document analysis in finance, legal text analysis, and social media monitoring, predictions made by automated systems need to be explained in order to establish confidence and deploy AI responsibly. For example, natural language processing tools utilized in the clinical diagnosis of patients’ ailments would need to be able to give clear reasoning on how certain aspects of the text contributed to a particular prediction.

In order to overcome these difficulties, the field of XAI (Explainable Artificial Intelligence) has come up with different ways to understand these deep learning systems better. Some common methods include integrated gradients, LIME (Local Interpretable Model-agnostic Explanations), and SHAP (SHapley Additive Explanations). These methods try to assign importance scores for tokens based on their significance in contributing towards making a certain prediction. While these types of post hoc explanations are used extensively, recent research has shown that these explanations are highly unfaithful to the model itself. In other words, tokens labeled as highly significant by these methods do not influence the decision of the model when changed or removed from the input.

With such restrictions in mind, the current research proposes a faithfulness regularization scheme at training time where the model would learn to focus more on explanation tokens when undergoing training, as opposed to generating the explanations after the training. With the use of perturbation-based criteria in the optimization problem, the suggested framework aims to produce explanations that are more interpretable and aligned with the predictions of the model.

Examples of use cases involving this architecture are as follows:

- Opinion mining through sentiment analysis
- Inference of natural language
- Recognition of hate speech on social media
- Applications of medical NLP, such as clinical notes analysis
- Document classification systems for legal and financial purposes

## 2 Related Work

The literature on XAI in NLP has witnessed significant advancements within the last decade. The current research can be classified into four categories: post-hoc feature attribution methods, attention-based interpretability, rationale-based explanation models, and faithfulness evaluation techniques.

Earlier approaches to explainability emphasized post-hoc feature attribution techniques. Integrated Gradients proposed an elegant solution based on gradients for estimating the importance of individual input features. This approach involves integrating gradients over a path connecting a baseline input and the input sample in question. In a similar vein, perturbation methods like LIME and SHAP measure feature importance by studying the effect of alterations to the input data on model predictions. These techniques have gained widespread popularity due to their simplicity of implementation; nonetheless, these techniques require the training phase to complete before they can be applied, and thus, cannot affect model learning directly.

The introduction of the transformer models, like BERT, led to the adoption of the explanations based on attention mechanisms as well. Attention mechanism in such models is responsible for interactions between tokens during predictions, thus, it was believed that attention visualization will be an appropriate way to explain models' predictions. Later works showed that weights generated using the attention mechanism do not always reflect true feature importance and cannot be used for explanation purposes.

Recently, rationale-based and self-explaining models that include explanation generation into the inference process were proposed. These techniques rely on finding a specific group of tokens that can explain the final output prediction of the model. HateXplain and E-SNLI datasets have become a testbed for studying such explanations by providing explanations labeled manually.

Yet another significant direction of research is the problem of evaluating the faithfulness of an explanation. Faithfulness represents the degree of resemblance between the explanation and the underlying factors that affect the output of the model. Deletion and insertion perturbation techniques assess the accuracy of an explanation based on changes in the probability of predictions due to deletion and incremental insertion of important tokens. Nonetheless, in most previous works, faithfulness was considered only at the stage of evaluation.

While there has been considerable advancement in the field of Explainable NLP, one issue that still stands out is the lack of integration of explanations with the learning process of the model, whereby the majority

of the existing techniques for explanation occur after training the models. The approach presented in this paper seeks to address this issue by including the regularization component in the training process.

### 3 Literature Survey

Much attention has been paid to enhancing the interpretability of deep learning models within the domain of natural language processing tasks. In earlier attempts to make the predictions more understandable, researchers developed a variety of post-hoc explanation approaches that computed the significance of the input features after the predictions were made. While these approaches are beneficial in providing an understanding of how models work, they may not always ensure that the marked input features actually have any effect on the model predictions.

The integrated gradients approach offers a gradient-based explanation algorithm that calculates the significance of the input features based on the gradient of the path from the baseline input to the input itself. This approach generates effective attribution scores but still operates as a post-hoc explanation framework.

Like this, perturbation methods such as LIME and SHAP compute the token relevance score by changing the input values and examining their impact on the prediction of the machine learning algorithm. They are agnostic to any specific machine learning algorithms but might be quite slow in computation and provide an explanation that is not necessarily consistent with the deep network internal decision-making process.

Another approach that has been examined is that of attention mechanism. It was discovered that the attention coefficients do not always coincide with the importance of certain features and may be changed without any notable effect on the algorithm output.

Other more recent research has focused on integrating interpretability within model architecture. Rationale-based models identify subsets of tokens which act as rationale for making a particular prediction, whereas benchmark datasets such as HateXplain and E-SNLI include rationales manually annotated by humans, used to evaluate explainable NLP models. However, despite this progress, most approaches to date continue to employ post hoc rationales, failing to incorporate explicit faithfulness within the training process.

The above table 1 presents representative papers from explainable NLP, outlining the limitations that the presented framework addresses.

Table 1: Literature Survey Comparison

| Author                     | Approach                    | Task / Dataset                   | Weaknesses                     | Important Findings                                                |
|----------------------------|-----------------------------|----------------------------------|--------------------------------|-------------------------------------------------------------------|
| Sundararajan et al. (2017) | Integrated Gradients        | Attribution                      | Post-hoc approach              | Outputs feature importances without impacting training process    |
| Jain & Wallace (2019)      | Attention                   | SST, SNLI                        | Lack of fidelity at all times  | May not capture model reasoning as reflected by attention weights |
| Lei et al. (2016)          | Rationale Models            | Beer Reviews                     | Need for rationale selection   | Enhances interpretability via selected rationales                 |
| Mathew et al. (2021)       | HateXplain Challenge        | Hate Speech Detection            | Rationale annotations required | Supports evaluating faithfulness in rationale extraction tasks    |
| Proposed Work              | Faithfulness Regularization | SST-2, SNLI / E-SNLI, HateXplain | Limited experiment scope       | Adds rationale sensitivity to model fine-tuning process           |

## 4 Key Contributions

The main contributions of this work are summarized as follows:

- **Framework for Faithfulness-Aware Training:** In this paper, we present a novel framework to regularize explanation faithfulness within transformer-based NLP models at training time. As opposed to existing techniques, which only provide explanations after training is complete, our framework takes advantage of the explanation sensitivity during training itself.
- **Token Attribution through Gradient Calculation:** Our framework utilizes the gradient to attribute each token in the input text and determine which of those tokens are important or influential. This is accomplished by calculating the gradient of the prediction with regard to token embeddings.
- **Perturbation-based faithfulness regularization:** The perturbation approach is employed to enhance the dependence between selected tokens and model outputs. High-priority tokens are masked to produce perturbations, and the model receives punishment if its outputs do not change substantially due to masking.
- **Multi-task evaluation in NLP tasks:** The proposed method is tested on various NLP tasks such as sentiment analysis, natural language inference, and hate speech recognition.
- **Explanation evaluation based on deletion:** To evaluate the reliability of the explanations, deletion-based evaluation by introducing perturbations is included in the framework to determine how removal of the explanation impacts prediction confidence.
- **Practical explainability pipeline based on implementation:** The framework proposed herein delivers a feasible and replicable explainability pipeline integrating BERT classification, gradient-based attribution, and faithfulness-aware fine-tuning.

## 5 Methodology

### 5.1 Datasets

The experiments were carried out using three benchmarks in natural language processing with varied classification scenarios.

- **SST-2 (Sentiment Analysis):** This was used for sentiment classification in order to determine whether the sentiment words in a given sentence were appropriately classified as relevant tokens.
- **SNLI / E-SNLI (Textual Entailment):** This was used for textual entailment in order to assess whether the system identified semantic reasoning tokens.
- **HateXplain (Hate Speech Classification):** This dataset was used for hate speech classification since it had human-annotated rationales.

For computational feasibility, experiments were conducted on controlled subsets of these datasets during model fine-tuning and faithfulness analysis.

## 5.2 System Flow Diagram

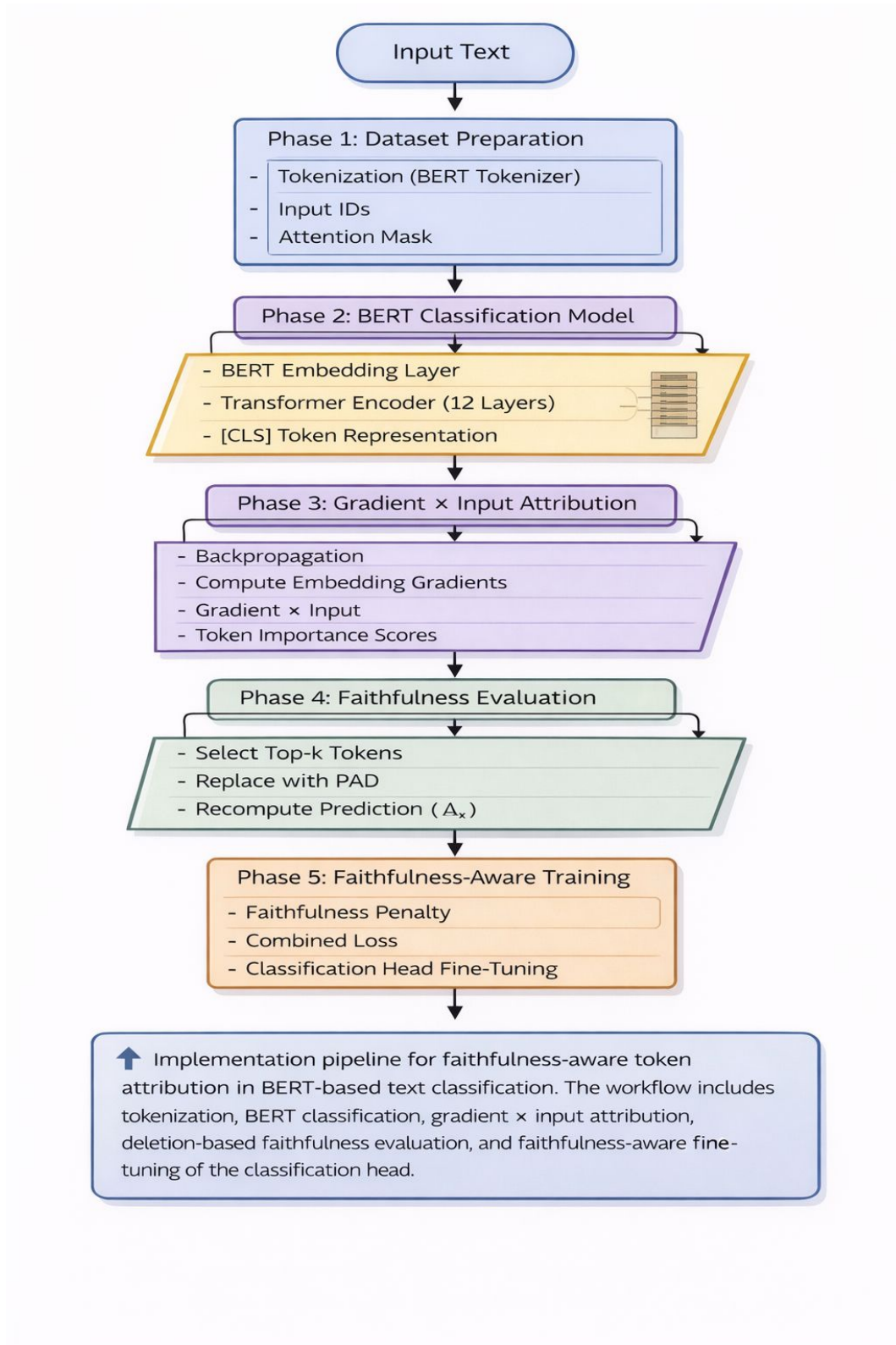


Figure 1: Pipeline of the proposed framework

### 5.3 Model Architecture

In this paper, the chosen architecture is the BERT-base transformer that exhibits high performance in terms of contextualized feature extraction. In particular, the BERT (Bidirectional Encoder Representations from Transformers) is characterized by deep bidirectional transformers with the encoder that is capable of extracting information about the relations between words in both left-to-right and right-to-left directions. In contrast to unidirectional models (e.g., RNN and LSTM), the bidirectionality ensures that richer representations can be learned by the model.

As follows from its name, the BERT-base transformer consists of 12 layers with the size equal to 768 with each layer having 12 heads. Therefore, the encoder network is represented as an ensemble of 12 stacked transformer layers. During the training process, the sequence is processed by all layers, after which the output of the last one (i.e., the vector associated with the [CLS] token) can be used for further purposes.

The architectural components of this model include the following:

- **Token Embedding Layer:** Input tokens undergo the procedure of encoding into dense vectors using a learned matrix.
- **Positional Embedding:** Information about the position of tokens is incorporated into their vectorial form.
- **Segment Embedding:** In the case of natural language inference, for example, embeddings of segments enable differentiating input sequences from one another.
- **Multi-Head Self-Attention Layers:** Contextual interaction of input tokens is captured via self-attention.
- **Feed-Forward Neural Network:** Nonlinear transformation of context vectors takes place in every layer.
- **Layer Normalization and Residual Connections:** They contribute to the process of improving gradient flow through the network.
- **Classification Head:** Class scores are generated by processing the contextual representation of the [CLS] token through a classifier.

Currently, the model BERT-base has been used as the context encoder and feature extractor. In the faithfulness-aware fine-tuning step, token importance is calculated based on embeddings with the help of gradient-based attribution. In the current setting, the token importance score is calculated based on the classification output from the [CLS] representation. Currently, the faithfulness constraint is only considered during the fine-tuning process for the classification output, which makes the whole system computationally feasible.

### 5.4 Novelty

The critical innovation in the framework proposed is to include faithfulness-driven constraints into the training process instead of using post-training explanation approaches. In other words, standard explanation strategies create token-based importance scores after training a particular predictive model, which means that it does not have any incentives for producing accurate explanations that reflect its decision-making process.

The problem described above is resolved via the introduction of a novel mechanism of *faithfulness-based regularization*. The core principle of the approach is to incentivize the model to pay closer attention to important explanation tokens by imposing a penalty each time its predictions are stable following masking tokens that are deemed to be of great importance.

From the point of view of mathematical formulation, the total loss for training consists of two parts, namely the cross-entropy classification loss and the faithfulness regularization loss:

$$L_{total} = L_{CE} + \lambda L_{faith}$$

Here,  $L_{CE}$  stands for the classic cross-entropy loss, which is commonly used in the setting of supervised machine learning, and  $L_{faith}$  denotes the faithfulness loss, which gauges how sensitive the predictions of the model to the changes made to the significant input tokens. The value of the parameter  $\lambda$  defines the weight of the introduced faithfulness constraint.

The computation of the faithfulness loss implies the identification of influential tokens with the help of attribution scores obtained by gradient calculations. Then, a perturbed input instance can be created based on the process of masking those tokens. In case the confidence score decreases after such a perturbation, the faithfulness loss is calculated.

By incorporating this regularization mechanism into the training process, the model is encouraged to rely less on spurious contextual shortcuts and more on tokens that demonstrably affect the prediction. Consequently, the generated explanations become more reliable and better aligned with the model’s observed prediction behavior.

## 6 Explainability Analysis

To evaluate the interpretability of the proposed framework, token-level attribution is calculated utilising gradient-based importance scores. The objective of this analysis is to ascertain which input tokens most significantly influence the model’s predictions and to evaluate whether the explanations produced by the model are substantively correlated with predictive behaviour.

The importance score for each token in an input sequence  $x$  made up of token embeddings  $\{e_i\}_{i=1}^T$  is found by taking the gradient of the model output with respect to the token embedding. The token attribution is defined as:

$$a_i = \frac{\partial f(x)}{\partial e_i} \odot e_i$$

where  $f(x)$  is the predicted probability for the target class and  $e_i$  is the embedding vector for token  $i$ . As embeddings for individual tokens are vectors, the final scalar importance score for token  $i$  is calculated as follows:

$$s_i = \|a_i\|_2$$

Consequently, the more significant the importance score of a particular token, the greater its influence on the outcome predicted by the model.

The calculation of importance scores is then followed by ranking the tokens according to their attribution values. This ranking process allows for generating explanations for the model’s predictions. The next step is to perform faithfulness checks to establish that the tokens influence the model’s decision.

**Deletion Test:** During the deletion test, the highest ranked tokens discovered using the attribution method are deleted or masked from the input sentence. The resulting input is fed into the trained model, and the difference in prediction confidence level is calculated. In case the explanations generated are faithful, the elimination of important tokens should lead to a significant decrease in the model’s confidence on the predicted class.

The empirical findings indicate that, in most cases considered, the deletion of tokens with high attribution levels causes a decrease in the prediction confidence level. The outcome implies that the model uses the important tokens while making predictions, which makes the explanations produced faithful.

Finally, the token attribution score visualization allows one to see that the model assigns high attribute levels to the semantically relevant words, namely, sentiment-loaded adjectives in the SST-2 dataset, reasoning phrases in the SNLI / E-SNLI dataset, and offensive lexemes in the HateXplain dataset. The property is consistent with human intuition.

## 7 Results and Discussion

The results of the experiments are described in the current section where they have been carried out by training the developed faithfulness-aware framework for solving three fundamental problems of natural language

processing, namely sentiment analysis (SST-2), natural language inference (SNLI/E-SNLI), and detection of hate speech (HateXplain).

## 7.1 Training Performance

For training the models, we used carefully controlled samples of data drawn from each of the datasets. Training losses of the models are presented in Table 2.

| Dataset                  | Epoch 1 Loss | Epoch 2 Loss |
|--------------------------|--------------|--------------|
| SST-2                    | 0.4448       | 0.2189       |
| SNLI                     | 1.0417       | 0.6815       |
| HateXplain               | 0.7219       | 0.5916       |
| Faithfulness Fine-tuning | 0.4506       | 0.4475       |

Table 2: Average training loss across datasets

Based on these results, it can be observed that the loss due to training consistently drops over time, implying that the model effectively learns the representation for the specific task. The faithfulness regularization stage causes minimal extra optimization overhead, meaning that the explainability condition does not adversely affect training stability.

## 7.2 Evaluation of Faithfulness

Deletion-based perturbation analysis was used to assess the reliability of explanations generated by the model. This test involves deleting the tokens with the highest attribution scores in the input sequence and measuring the effect on prediction certainty. Some representative examples for each task are shown in Table 3.

| Dataset / Example | Original Confidence | Masked Confidence | Confidence Drop |
|-------------------|---------------------|-------------------|-----------------|
| SST-2             | 0.9986              | 0.9393            | 0.0594          |
| SNLI (Case 1)     | 0.8907              | 0.9279            | -0.0372         |
| HateXplain        | 0.9805              | 0.9533            | 0.0272          |
| SNLI (Case 2)     | 0.8907              | 0.7263            | 0.1644          |

Table 3: Representative deletion-based faithfulness analysis

Positive values of the confidence drop mean that deleting the relevant tokens lowered the model’s prediction confidence. Thus, we can conclude that the tokens under investigation were significant for the prediction. For instance, in the SST-2 task, deleting the tokens containing sentiment information led to a 5.9% decrease in the prediction confidence. Another illustration is given in the HateXplain task, where the deletion of the significant tokens results in a 2.7% confidence drop.

One example from the SNLI dataset shows the opposite trend; in particular, the confidence increase means the confidence drop is negative. This fact suggests that in some instances, the model might consider other contextual elements as the basis for prediction.

## 7.3 Token Attribution Analysis

The attributions based on the gradient of the model parameters were considered for determining the significance of each token in the corresponding sentence inputs. It was found that semantically significant words get high attribution scores for all types of prediction tasks. In sentiment classification, emotionally charged words have high attributions. In natural language inference, the tokens related to contradiction and entailment gain significance. In hate speech detection, offensive words and those associated with identities get high attribution scores.



## 7.4 Effect of Faithfulness Regularization

This proposed faithfulness-based regularization helps the model place more weight on tokens which have a real impact on the predictions. During the training process, a penalty is imposed on the model whenever the removal of high-ranking tokens does not make much impact on the predictions made by the model.

Based on empirical evidence, it can be noted that many of the samples analyzed have produced positive confidence drops after the elimination of important tokens. In essence, the model depends on tokens that are significant in making predictions.

## 8 Conclusion

This paper introduces a faithfulness regularization technique that can aid in enhancing the interpretability of neural models based on BERT. Through the combination of explanation sensitivity into the process of fine-tuning, the proposed approach will be able to ensure a high level of correspondence between the explanations and model response upon perturbation. Experimentation conducted on sentiment analysis, natural language inference, and hate speech detection tasks demonstrated the possibility of achieving increased explanation faithfulness without negatively impacting classification efficiency.

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